

WHEN SIZE MATTERS MOST!

Over the years, you will have come across the word nanotechnology, in Digit. In fact, we see it used so often, in newspapers, on TV and on the web – as if it were a common, everyday word. However, although a lot of us will know “nano” means “small” – some will have Tata Motors to thank for that – most of us don’t know anything beyond that about the massive world that exists at the miniscule atomic level. When asked, many will say, “It’s the technology of the future”, and will even cite articles they read, and use words such as nanobots, or buckyballs in sentences. However, we aim to improve that by giving you some comprehensive knowledge about what is perhaps the most exciting field of science in our era. Nanotechnology does, after all, promise to be the fountain of youth, the elixir of life, the harbinger of death and the end of mankind, all rolled into one.

We’ll go small, really small – down to atomic levels. We’ll take a look at nanomaterials, and how hard it is to see them, let alone manufacture them. We’ll look at the breakthroughs that gave birth to this science, and take a look at where we’re headed. What will a nanotech future hold in store for us? Will nanotechnology bring peace on Earth, one way or another – bring the end of terrorism, or wipe out humanity? Will we even stay human, some ask, if we’ve got nanobots scampering around inside us, fixing everything that breaks, transforming us into cyborgs?

We will try and answer all of these questions, and more, through the course of this book. We’ll empower you with knowledge, and you can make up your own minds whether nanotechnology is really a boon or a bane. As always, we look forward to your feedback on the topic. 